Topic: Spring and mass systems

Question: A mass of weight 50 pounds stretches a spring 10 feet. Find the equation that models the motion of the mass if we release the mass when $t = 0$ from a position of 2 feet above equilibrium, with a downward velocity of 5 ft/s.

Answer choices:

A $x(t) = -2\cos\left(\frac{4\sqrt{5}}{5}t\right) + \frac{5\sqrt{5}}{4}\sin\left(\frac{4\sqrt{5}}{5}t\right)$

B $x(t) = 2\cos\left(\frac{\sqrt{10}}{10}t\right) - \sin\left(\frac{\sqrt{10}}{10}t\right)$

C $x(t) = \cos\left(\frac{\sqrt{10}}{10}t\right) - \sin\left(\frac{\sqrt{10}}{10}t\right)$

D $x(t) = 2\cos\left(\frac{4\sqrt{5}}{5}t\right) - \frac{5\sqrt{5}}{4}\sin\left(\frac{4\sqrt{5}}{5}t\right)$

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Solution: A

First we'll use Hooke's Law to find the spring constant k.

$$F = ks$$

$$50 = k(10)$$

$$k = 5 \text{ lb/ft}$$

To convert weight into mass, we'll use $m = W/g = 50/32 = 25/16$ slug. Plugging everything we have into the second order equation, we get

$$\frac{25}{16}\left(\frac{d^2x}{dt^2}\right) + 5x = 0$$

$$\frac{d^2x}{dt^2} + \frac{16}{5}x = 0$$

From this equation, we see that $\omega = \sqrt{16/5} = 4/\sqrt{5}$, which means that the general solution is given by

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$$x(t) = c_1 \cos\left(\frac{4}{\sqrt{5}}t\right) + c_2 \sin\left(\frac{4}{\sqrt{5}}t\right)$$

Its derivative is

$$x'(t) = -\frac{4}{\sqrt{5}}c_1 \sin\left(\frac{4}{\sqrt{5}}t\right) + \frac{4}{\sqrt{5}}c_2 \cos\left(\frac{4}{\sqrt{5}}t\right)$$

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The question states that the initial position is $x(0) = -2$ and the initial velocity is $x'(0) = 5$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$-2 = c_1 \cos\left(\frac{4}{\sqrt{5}}(0)\right) + c_2 \sin\left(\frac{4}{\sqrt{5}}(0)\right)$$

$$-2 = c_1(1) + c_2(0)$$

$$c_1 = -2$$

and

$$5 = -\frac{4}{\sqrt{5}}c_1 \sin\left(\frac{4}{\sqrt{5}}(0)\right) + \frac{4}{\sqrt{5}}c_2 \cos\left(\frac{4}{\sqrt{5}}(0)\right)$$

$$5 = -\frac{4}{\sqrt{5}}c_1(0) + \frac{4}{\sqrt{5}}c_2(1)$$

$$c_2 = \frac{5\sqrt{5}}{4}$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = -2\cos\left(\frac{4}{\sqrt{5}}t\right) + \frac{5\sqrt{5}}{4}\sin\left(\frac{4}{\sqrt{5}}t\right)$$

$$x(t) = -2\cos\left(\frac{4\sqrt{5}}{5}t\right) + \frac{5\sqrt{5}}{4}\sin\left(\frac{4\sqrt{5}}{5}t\right)$$

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Topic: Spring and mass systems

Question: A mass weighing 10 pounds stretches a spring 2 feet. Find the equation that models the motion of the mass if we release it when $t = 0$ from a position of 5 feet above equilibrium, with a downward velocity of 1 ft/s.

Answer choices:

A $x(t) = 5\cos\left(\frac{\sqrt{2}}{2}t\right) - 2\sin\left(\frac{\sqrt{2}}{2}t\right)$

B $x(t) = 5\cos\left(\frac{\sqrt{2}}{2}t\right) + 2\sin\left(\frac{\sqrt{2}}{2}t\right)$

C $x(t) = -5\cos(4t) - \frac{1}{4}\sin(4t)$

D $x(t) = -5\cos(4t) + \frac{1}{4}\sin(4t)$

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Solution: D

First we'll use Hooke's Law to find the spring constant k.

$$F = ks$$

$$10 = k(2)$$

$$k = 5 \text{ lb/ft}$$

To convert weight into mass, we'll use $m = W/g = 10/32 = 5/16$ slug. Plugging everything we have into the second order equation, we get

$$\frac{5}{16}\left(\frac{d^2x}{dt^2}\right) + 5x = 0$$

$$\frac{d^2x}{dt^2} + 16x = 0$$

From this equation, we see that $\omega = \sqrt{16} = 4$, which means that the general solution is given by

$$x(t) = c_1 \cos(\omega t) + c_2 \sin(\omega t)$$

$$x(t) = c_1 \cos(4t) + c_2 \sin(4t)$$

Its derivative is

$$x'(t) = -4c_1 \sin(4t) + 4c_2 \cos(4t)$$

The question states that the initial position is $x(0) = -5$ and the initial velocity is $x'(0) = 1$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

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$$-5 = c_1 \cos(4(0)) + c_2 \sin(4(0))$$

$$-5 = c_1(1) + c_2(0)$$

$$c_1 = -5$$

and

$$1 = -4c_1 \sin(4(0)) + 4c_2 \cos(4(0))$$

$$1 = -4c_1(0) + 4c_2(1)$$

$$c_2 = \frac{1}{4}$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = -5\cos(4t) + \frac{1}{4}\sin(4t)$$

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Topic: Spring and mass systems

Question: A mass weighing 16 pounds stretches a spring 8 feet, and a damping force that's triple the instantaneous velocity is acting on the spring/mass system. Find an equation that models the motion of the mass if it's initially released from 4 feet below equilibrium with an upward velocity of 6 ft/s.

Answer choices:

A $x(t) = e^{3t}((3+\sqrt{5})e^{\sqrt{5}t} + (3-\sqrt{5})e^{-\sqrt{5}t})$

B $x(t) = e^{-3t}((-3+\sqrt{5})e^{\sqrt{5}t} + (-3-\sqrt{5})e^{-\sqrt{5}t})$

C $$x(t) = e^{3t}\left(\left(1-\frac{\sqrt{5}}{2}\right)e^{\sqrt{5}t} + \left(-5-\frac{\sqrt{5}}{2}\right)e^{-\sqrt{5}t}\right)$$

D $$x(t) = e^{-3t}\left(\left(2+\frac{3\sqrt{5}}{5}\right)e^{\sqrt{5}t} + \left(2-\frac{3\sqrt{5}}{5}\right)e^{-\sqrt{5}t}\right)$$

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Solution: D

Hooke's Law tells us that the spring constant is

$$F = ks$$

$$16 = k(8)$$

$$k = 2 \text{ lb/ft}$$

Now we'll use $W = mg$ to convert the weight into mass.

$$W = mg$$

$$16 = m(32)$$

$$m = \frac{1}{2} \text{ slug}$$

With the spring constant and the mass, and $\beta = 3$ since the damping force is triple the instantaneous velocity, we can plug everything into the differential equation.

$$\frac{d^2x}{dt^2} + \frac{\beta}{m}\left(\frac{dx}{dt}\right) + \frac{k}{m}x = 0$$

$$\frac{d^2x}{dt^2} + \frac{3}{\frac{1}{2}}\left(\frac{dx}{dt}\right) + \frac{2}{\frac{1}{2}}x = 0$$

$$\frac{d^2x}{dt^2} + 6\left(\frac{dx}{dt}\right) + 4x = 0$$

The associated characteristic equation and its roots are

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$$r^2 + 6r + 4 = 0$$

$$r = -3 \pm \sqrt{5}$$

Because we find roots where $\lambda^2 - \omega^2 > 0$, the spring and mass system is overdamped and the general solution is

$$x(t) = e^{-\lambda t}(c_1 e^{\sqrt{\lambda^2 - \omega^2}t} + c_2 e^{-\sqrt{\lambda^2 - \omega^2}t})$$

$$x(t) = e^{-3t}(c_1 e^{\sqrt{5}t} + c_2 e^{-\sqrt{5}t})$$

Its derivative is

$$x'(t) = -3e^{-3t}(c_1 e^{\sqrt{5}t} + c_2 e^{-\sqrt{5}t}) + e^{-3t}(\sqrt{5}c_1 e^{\sqrt{5}t} - \sqrt{5}c_2 e^{-\sqrt{5}t})$$

$$x'(t) = e^{-3t}((-3 + \sqrt{5})c_1 e^{\sqrt{5}t} - (3 + \sqrt{5})c_2 e^{-\sqrt{5}t})$$

The question states that the initial position is $x(0) = 4$ and the initial velocity is $x'(0) = -6$, so we'll plug these into $x(t)$ and $x'(t)$, and we get

$$4 = e^{-3(0)}(c_1 e^{\sqrt{5}(0)} + c_2 e^{-\sqrt{5}(0)})$$

$$4 = c_1 + c_2$$

$$c_1 = 4 - c_2$$

and

$$-6 = e^{-3(0)}((-3 + \sqrt{5})c_1 e^{\sqrt{5}(0)} - (3 + \sqrt{5})c_2 e^{-\sqrt{5}(0)})$$

$$-6 = (-3 + \sqrt{5})c_1 - (3 + \sqrt{5})c_2$$

$$-6 = (-3 + \sqrt{5})(4 - c_2) - (3 + \sqrt{5})c_2$$

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$$-6 = 4(-3+\sqrt{5}) - (-3+\sqrt{5})c_2 - (3+\sqrt{5})c_2$$

$$-6 = -12 + 4\sqrt{5} + (3-\sqrt{5}-3-\sqrt{5})c_2$$

$$6 - 4\sqrt{5} = -2\sqrt{5}c_2$$

$$c_2 = 2 - \frac{3\sqrt{5}}{5}$$

Then the value of c_1 is

$$c_1 = 4 - c_2$$

$$c_1 = 4 - \left(2 - \frac{3\sqrt{5}}{5}\right)$$

$$c_1 = 2 + \frac{3\sqrt{5}}{5}$$

So the equation modeling the motion of this spring and mass system, with these particular initial conditions for velocity and position, is given by

$$x(t) = e^{-3t}\left(\left(2 + \frac{3\sqrt{5}}{5}\right)e^{\sqrt{5}t} + \left(2 - \frac{3\sqrt{5}}{5}\right)e^{-\sqrt{5}t}\right)$$

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